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Points-supervised Fundus Vessel Segmentation via Shape Priors and Contrastive Learning

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***Abstract*—The performance of fully supervised methods for fundus vessel segmentation highly relies on a large number of full labels which are laborious and time-consuming to obtain. Although weak annotations relax the requirement for pixel-wise labeling, they pose challenges in learning comprehensive information about the target. Some methods use pseudo labels generated from network predictions for extra supervision, but false positive predictions in these labels may harm training. In this paper, to tackle this problem and to balance the annotation cost and supervision information, we introduce point annotations to fundus vessel segmentation and propose a novel method, called Points-based Vessel segmentation Network (PVN), to enhance the segmentation accuracy. In PVN, to avoid noise in pseudo labels, by combining proposed Point Activation Maps, shape priors of vessels are learned and used as soft supervision. Additionally, to further leverage the annotated vessel and background points, we design a novel contrastive learning method in a pixels-and-regions-mixed manner, which helps learn discriminative features by distinguishing between pixel and region samples of vessels and background. The performance of PVN is evaluated on laser speckle contrast imaging fundus images, 548 nm fundus images, and three public datasets, where PVN outperforms other point-supervised methods. Even with only 1% annotated pixels, PVN still achieves excellent performance. Our method is also flexible and easy to be combined with other frameworks. To the best of our knowledge, we are the first to propose and demonstrate the effectiveness of point annotations for fundus vessel segmentation. Our code is available at: https://github.com/kaiwenli325/PVN.**

***Index Terms*—Fundus vessel segmentation, laser speckle contrast imaging, point annotations, shape priors, contrastive learning.**

I. INTRODUCTION

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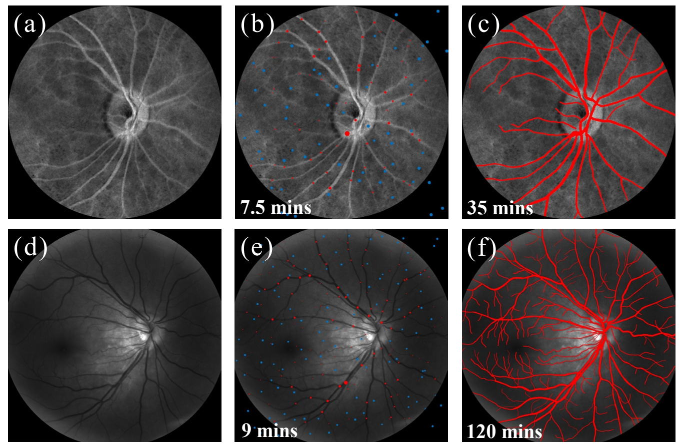

Fig. 1. Examples of point annotations versus full annotations: (a) An example of LSCI images. (b) Point annotations for (a). (c) Full annotations for (a). (d) An example of 548 nm fundus images. (e) Point annotations for (d). (f) Full annotations for (d). (mins: minutes).

FUNDUS vessel segmentation is a foundational task for many clinical applications like vessel wall thickness calculation [1]. Different modalities are available for imaging fundus vessels, including laser speckle contrast imaging (LSCI) [2] and multispectral imaging (MSI) [3]. LSCI obtains image contrast from the changes of speckle pattern caused by moving blood cells [2], as shown in Fig. 1(a). By imaging fundus at different wavelengths, MSI shows the spectral absorption characteristics of hemoglobin, where arteries and veins can both be visualized clearly under 548 nm [3], as shown in Fig. 1(d). With precise vessel segmentation results, oxygenation parameters can be calculated for each vessel in MSI images, and LSCI can measure blood flow parameters and vascular metrics [1]. Based on these measurements, diseases including hypertension and diabetes [1] can be better understood and diagnosed.

Although fully supervised methods for fundus vessel segmentation [4] can produce accurate results, the annotating process of numerous full labels is tedious and time-consuming, especially for images with many tiny vessels and strong noise. While proposed to reduce vessel labeling efforts, some works [5]–[8] still require several fully annotated masks or complex manual intervention. Recently, to alleviate the labeling burden, different forms of weak annotations have been proposed, including image-level labels [9], bounding boxes [10], scribbles

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[11] and points [12]. Image-level labels are mainly used to train an image classification network, from which the Class Activation Map (CAM) [13] is generated to get coarse localization of objects as pseudo segmentation masks. However, in fundus image datasets, each image contains vessels and background, which makes image-level labels unsuitable. Bounding boxes cannot effectively annotate mesh-like and fine targets like vessels. Scribbles require extra labeling efforts to accurately trace the shape of vessels, especially for small ones. In contrast, point annotations balance annotation efforts and supervision information, which can also be easily applied to large-scale datasets. Meanwhile, in further annotation of arteries and veins, points can be easily applied to the intersecting regions. Therefore, we introduce point annotations into vessel segmentation. As shown in Fig. 1, based on point annotations where each point contains a few pixels in a circle shape for vessels and background, the labeling time is significantly reduced from 35 minutes to 7.5 minutes for LSCI images and from 120 minutes to 9 minutes for 548 nm fundus images.

However, point annotations lack the information of full vessel extent, and pseudo labels generated from network predictions contain false positive predictions which may impair the segmentation performance. Thus, to address these issues and seek additional supervision, our proposed **P**oints-based **V**essel segmentation **N**etwork (PVN) utilizes the estimated vessel area as prior knowledge for soft supervision, inspired by [14]. As a fundamental component for this task and inspired by CAM, Point Activation Map (PAM) was proposed to generate coarse localization maps of targets by weighted summation of different feature maps. Specifically, in PVN, PAMs from different network layers are integrated to obtain accurately estimated area, where the weights in PAM for different feature maps are calculated only by the gradients from labeled vessel pixels. To avoid training noise, PVN also dynamically updates the estimated vessel area with momentum. Instead of directly using PAMs from the segmentation model which converges slowly with several training objectives, an auxiliary model is used for its faster convergence and thus gives more accurate estimation at an early training stage.

Meanwhile, only supervised by sparse points, it is hard for networks to distinguish between vessels and background in low-contrast and noisy regions, since there is no sufficient guidance to help networks learn discriminative features. Therefore, to further utilize the limited supervision provided by labeled points, supervised contrastive learning is used, which helps learn representative features by pulling samples of the same class close and by pushing samples of different classes apart. In PVN, contrastive learning is implemented between vessels and background in a pixels-and-regions-mixed manner. Specifically, beyond the comparisons between pixels from annotated points, region features of annotated points are generated by averaging the embeddings of pixels in each point, to compensate for the lack of image content in pixel-level comparisons. Compared with averaging all pixels of each class in an image as region embeddings [15], [16], our point-based region features preserve the variability of vessels and background at different locations. Additionally, during loss calculation, position-based weights are assigned to labeled vessel pixels, which emphasizes edge pixels to bring more supervision on vessel boundaries since sparse annotations do not contain fine boundary information.

In a nutshell, our contributions are fourfold:

- To the best of our knowledge, this work is the first to introduce point annotations to vessel segmentation and demonstrate its efficiency, which greatly reduces labeling burden. Our private datasets will be public.
- A novel shape priors-based soft supervision method is proposed to compensate for the limited supervision in point annotations.
- A pixels-and-regions-mixed contrastive learning is proposed with weighing approaches suitable for point annotations.
- Experiments indicate that our method can effectively segment vessels with point labels and can be easily integrated with existing segmentation models.

II. Related works

A. Vessel segmentation

Although fully supervised deep learning methods have achieved great success in vessel segmentation [4], the complex annotation process hinders its further application. Recently, to deal with this problem, several semi-supervised [5]–[8] and weakly supervised [11], [17]–[19] methods have been proposed. For instance, Hou et al. [5] used labeled, unlabeled, and synthetic images from a GAN generator to train a discriminator for per-pixel classification, where features of the generator were leaked as perturbations to the discriminator to improve the learning process. For unlabeled data, Hu et al. [6] designed a separate vessel skeleton prediction model, whose results were compared with the skeleton derived from the predicted masks as additional supervision. In [7], a hybrid level set model was used to generate pseudo masks. Yang et al. [8] manually labeled a subset of images, and for the others, they used an HSV color detection method to produce pseudo masks which were iteratively updated in the training process. Chen et al. [17] used CycleGAN [20] to generate synthetic LSCI images from the full labels of DRIVE dataset [21] for training a UNet. Different from [17], Yao et al. [19] took images and their pseudo labels generated by threshold segmentation as two image domains for CycleGAN, and they directly used one generator as the segmentor. In [18], multiple pseudo masks for each image were generated with different HSV thresholds, where the best mask was selected manually as the final label. Chinkamol et al. [11] generated scribbles from full masks as weak supervision. Although these works help reduce annotation labor, the fully labeled masks used in semi-supervised methods still require expert knowledge. Furthermore, supervision obtained from pseudo labels or GAN-generated images is prone to false positives and domain shifts.

B. Learning from point annotations

As proposed in [12], point annotations have been applied for both semantic and instance segmentation on natural and medical images, where methods were mainly based on pseudo

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masks [22]–[27], regularized losses [14], [28], auxiliary supervision [12], [29], and contrastive learning [30], [31]. First, Zhao et al. [27] iteratively updated pseudo masks from network predictions using self-training and co-training. Chen et al. [22] initially assigned pseudo labels to superpixels according to annotated points and then expanded pseudo labels to other superpixels according to feature similarity. Tree energy loss [24] used tree filter to model image color-based and feature-based affinities which were applied to generate pseudo labels from the model predictions. Moreover, variational models [26] and the paradigm of optimal transport [23] were also utilized for pseudo mask generation. As for regularized losses, CRF loss [28] took image content and dependencies of affinity into account. SizeLoss [14] incorporated priors of target shape by constraining the area of predicted targets. However, the performance of SizeLoss heavily relied on the area calculated from the full masks. Auxiliary supervision is another approach used in point-supervised segmentation. Bearman et al. [12] predicted each pixel's probability of belonging to an object and combined it in the loss function to encourage networks to place importance on pixels with high probability. PseudoEdgeNet [29] introduced an extra loss item by comparing the target boundaries generated by a shallow network with those from the predicted masks. Considering the distribution of nuclei in histopathology images, Qu et al. [25] used Voronoi edges, which separated the nuclei, as extra supervision of background in the loss function. Finally, it is natural to adopt contrastive learning to incorporate comparisons between annotated points of different classes. Lee et al. [31] sampled uncertain points into the memory bank to deal with the problem of small objects being easily ignored. Ke et al. [30] focused on establishing more contrastive relationships between pixels. However, there is no research for points-based vessel segmentation, where greater efforts for full annotations are required.

C. Contrastive learning

Contrastive learning enhances the representation of features by grouping features of the same class together and separating features of different classes further apart. In this process, a specific pixel serves as a reference (anchor) to identify pixels of the same class (positive/pos samples) and those of different classes (negative/neg samples). Although initially proposed for unsupervised learning [32], contrastive learning has been gradually applied in a supervised manner [15], [16]. Viewing contrastive learning as a dictionary look-up problem, MoCo [33] used a momentum encoder to get consistent features of images and memory banks to store sufficient samples, which were proven to be two key factors for contrastive learning. In contrast, SimCLR [32] used large minibatches to provide sufficient samples and showed the importance of more data augmentation and nonlinear transformation as the projection head. In the supervised scenario, contrastive loss mainly cooperates with other supervised losses, as an auxiliary approach to mine context. For example, in [15], region centers, generated by weighted average of features from the same class, were used as anchors and pos/neg samples in contrastive learning. Whereas in [16], different sampling methods of anchors and pos/neg samples were compared. Unlike the abovementioned methods, we use anchor sampling and memory bank updating methods at the level of both pixels and independent point regions with weighting methods designed for points, which can fully exploit annotations.

III. Methods

In this section, we describe the proposed PVN in detail. First, we present the overall architecture of PVN. Then, the shape priors-based soft supervision for vessels is introduced. Afterward, we explain the contrastive learning between vessels and background. Finally, we provide the complete loss function.

A. Method Overview

As illustrated in Fig. 2, the proposed PVN consists of three main components: a segmentation model $\mathcal{M}_{seg}$, a soft supervision generation process, and a contrastive learning part. Given an input image $I \in \mathbb{R}^{\mathrm{C\times H\times W}}$ in a minibatch $B \in \mathbb{R}^{\mathrm{B\times C\times H\times W}}$, where B, C, H and W represent the batch size, number of color channels, image height and image width, the feature extractor F_{cnn} and the segmentation head F_s of $\mathcal{M}_{seg}$ produce the predicted logits $\mathcal{Y}^L = F_s(F_{cnn}(I)) \in \mathbb{R}^{\mathrm{K\times H\times W}}$. Then, softmax is applied to generate the segmentation map $\hat{Y} = \mathrm{softmax}(\mathcal{Y}^L) \in [0,1]^{\mathrm{K\times H\times W}}$, representing the probabilities that each pixel belongs to one of K classes (in our case, $\mathrm{K}=2$). For labeled pixels in I, the partial cross-entropy loss, $\mathcal{L}_{pce}$, is directly used, which is defined as:

$$\mathcal{L}_{pce} = -\sum_{j}^{|\Omega_P|}\sum_{k}^{\mathrm{K}} a_k Y_j^k \log(\hat{Y}_j^k) \tag{1}$$

where $Y \in \{0,1\}^{\mathrm{K\times H\times W}}$ is the ground truth label, and Ω_P is the collection of labeled pixels. To address class imbalance, a weighting factor a_k is introduced:

$$a_k = 1 - \frac{|\Omega_P^k|}{|\Omega_P|} \tag{2}$$

where Ω_P^k is the collection of annotated pixels of class k.

In the soft supervision generation part, learned vessel shape priors are used to limit the predicted shape of vessels in $\mathcal{L}_{area}$. Meanwhile, the pixels-and-regions-mixed contrastive learning encourages $\mathcal{M}_{seg}$ to fully leverage the annotation information in $\mathcal{L}_{cl}$. The implementation of $\mathcal{L}_{area}$ and $\mathcal{L}_{cl}$ are detailed in the following sections. After the training is completed, only $\mathcal{M}_{seg}$ is used for inference.

B. Soft Supervision Generation

In PVN, learned prior knowledge of vessel area serves as additional soft supervision. Firstly, PAMs from an auxiliary model are used for area estimation. Specifically, as shown in Fig. 2(A), the auxiliary model ($\mathcal{M}_{aux}$) shares the same F_{cnn} structure as $\mathcal{M}_{seg}$, but only trained by the partial cross-entropy loss. The logits predicted by $\mathcal{M}_{aux}$ for vessels in I are denoted as $\mathcal{Y}_v^{Laux} \in \mathbb{R}^{\mathrm{H\times W}}$, and the sum of $\mathcal{Y}_v^{Laux}$ for annotated vessel pixels is $\sum_{j\in\Omega_P^v}\mathcal{Y}_{vj}^{Laux}$, where Ω_P^v is the collection of

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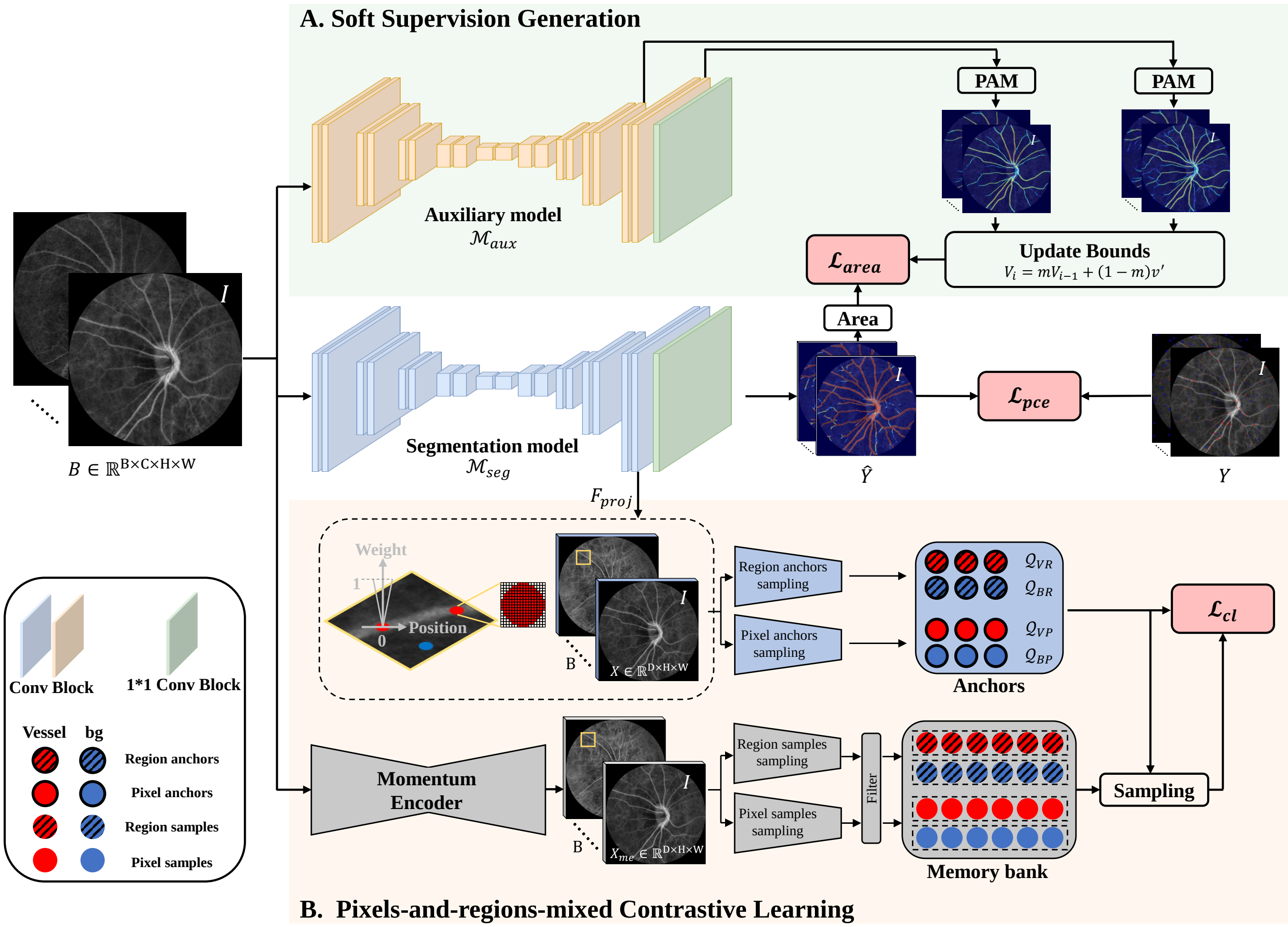

Fig. 2. Proposed PVN for point annotations-based fundus vessel segmentation: A. The shape priors-based soft supervision generation process for limiting the area of predicted vessels, where shape priors are learned from PAMs of different decoder layers. B. Contrastive Learning using pos/neg samples and anchors sampled from annotated pixels and points. Along with the anchor sampling, position-based weights are used for each pixel embedding.

labeled vessel pixels. Following Grad-CAM [13], the gradients of $\sum_{j\in\Omega_P^v}\mathcal{Y}_{vj}^{L_{aux}}$ to all pixels in a feature map are computed and averaged as the weight of this feature map, and the same process is repeated for all feature maps in a convolution layer. After using these weights to sum all feature maps and using ReLU to exclude negatives, a PAM map is generated for this convolution layer:

$$\mathrm{PAM} = \mathrm{ReLU}\Big(\sum_{n}^{\mathrm{N}'}\frac{1}{\mathrm{H}'\cdot\mathrm{W}'}\Big(\sum_{h}^{\mathrm{H}'}\sum_{w}^{\mathrm{W}'}\frac{\partial(\sum_{j\in\Omega_P^v}\mathcal{Y}_{vj}^{L_{aux}})}{\partial A_{h,w}^n}\Big)\cdot A^n\Big) \quad (3)$$

where $A_{h,w}^n$ is the n-th feature map in the convolution layer, H' is its height, W' is its width, and N' is the number of feature maps in this layer. As illustrated in Fig. 3(a) and (b), the regions that positively contribute to the predicted vessels are highlighted, which can be viewed as the coarse localization of vessels.

However, PAMs from different convolution layers exhibit different levels of granularity of vessel localization. As illustrated in Fig. 3, the coarse localization map (Fig. 3(a)) from a deeper decoder layer in UNet can detect more vessels.

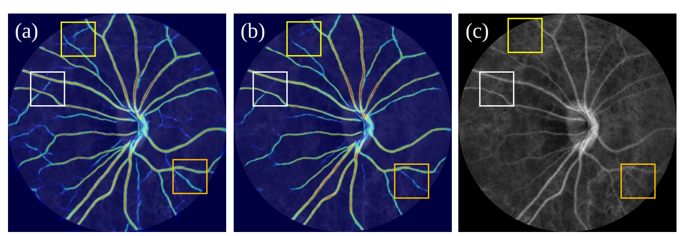

Fig. 3. Comparisons of PAMs from different UNet decoder layers after warmup of 20 epoches: (a) PAM generated by the features of the second-to-last decoder layer in UNet. (b) PAM generated by the features of the last decoder layer in UNet. (c) The original image. Rectangles of different colors highlight the regions for comparison.

In contrast, although the map (Fig. 3(b)) from a shallower decoder layer in UNet has more precise vessel boundaries, it also misses vessels in low-contrast regions. Therefore, we average the area of vessels in PAMs from different layers to get more accurate area estimation. Formally, for each iteration, the vessel area of I in different PAMs from layer l to layer $l+n$ of $\mathcal{M}_{aux}$ is denoted as v_l, v_{l+1},..., v_{l+n} and the averaged area of them is v'. To avoid training noise, the current iteration's

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vessel area V_i is calculated by a momentum update with the area from previous iterations V_{i-1}:

$$V_i = mV_{i-1} + (1 - m)v' \tag{4}$$

where i represents the iteration index.

After obtaining estimated area, soft supervision is made on the predictions of $\mathcal{M}_{seg}$:

$$\mathcal{L}_{area} = \begin{cases} \left(V_{pi} - b_i^{upper}\right)^2, & V_{pi} > b_i^{upper} \\ \left(V_{pi} - b_i^{lower}\right)^2, & V_{pi} < b_i^{lower} \\ 0, & otherwise \end{cases} \tag{5}$$

where V_{pi} means the predicted vessel area of $\mathcal{M}_{seg}$ in the i-th iteration, and b_i^{upper} and b_i^{lower} are defined as:

$$\begin{cases} b_i^{upper} = \alpha^{upper} \times V_i \\ b_i^{lower} = \alpha^{lower} \times V_i \end{cases} \tag{6}$$

where α^{upper} and α^{lower} are used to set the constraint range.

C. Pixels-and-regions-mixed Contrastive Learning

Distinguishing between vessels and background in regions with low signal-to-noise ratio requires representative features which are hard to obtain without the knowledge of full extent of vessels under point-supervised scenarios. Thus, to form a more discriminative feature space, PVN utilizes contrastive learning which reduces the distance between anchors and their positive samples while increasing the distance from negative samples, where positive samples belong to the same category as the anchors and negative samples are from a different category.

Formally, in PVN, infoNCE [34] is used as the contrastive learning loss $\mathcal{L}_{cl}$. For each anchor q_i in the anchor set $\mathcal{Q}$:

$$\mathcal{L}_{cl}^i = \frac{1}{\left|\mathcal{K}_i^+\right|}\sum_{k^+} -\omega_i \log \frac{\exp(q_i \cdot k^+/\tau)}{\exp(q_i \cdot k^+/\tau) + \sum_{k^-}\exp(q_i \cdot k^-/\tau)} \tag{7}$$

where $k^+ \in \mathcal{K}_i^+$, $k^- \in \mathcal{K}_i^-$, and τ is the temperature parameter [35]. $\mathcal{K}_i^+$ and $\mathcal{K}_i^-$ contain embeddings of positive and negative samples of q_i, respectively. For each q_i, half samples of $\mathcal{K}_i^+$ are selected by hard sampling from a memory bank $\mathcal{K}$ containing numerous samples, where k^+ with inner product $q_i \cdot k^+$ closer to -1 is a hard sample. For $\mathcal{K}_i^-$, samples with $q_i \cdot k^-$ closer to 1 are hard ones. The other half of $\mathcal{K}_i^+$ and $\mathcal{K}_i^-$ are sampled randomly. ω_i is a weight which will be introduced later.

However, solely relying on sparsely sampled pixels will neglect the contribution of regional image context in shaping the feature space. In this study, each annotated point is a small circular region containing several pixels. For an image I, the region set $\Omega_R = \left\{\mathcal{R}_n^k | k \in \{\mathrm{v}, \mathrm{b}\}; n = 1, 2, ...\right\}$ is the collection of labeled points and $\mathcal{R}_n^k = \left\{p_{ni}^k | i = 1, 2, ...\right\}$ where p_{ni}^k is the labeled pixels, v stands for "vessel" and b means "background". Meanwhile, the pixel set $\Omega_P = \left\{p_{ni}^k | k \in \{\mathrm{v}, \mathrm{b}\}; n = 1, 2, ...; i = 1, 2, ...\right\}$. Therefore, in PVN, treating each $\mathcal{R}_n^k$ as a region, we implement $\mathcal{L}_{cl}$ in a pixels-and-regions-mixed scheme between annotated vessel and background pixels and points in 1) anchor sampling to build $\mathcal{Q}$ and 2) memory bank $\mathcal{K}$ update. At the same time, various weighting and masking methods are applied to avoid the training noise and to emphasize the vessel boundaries.

1) Anchor sampling: For a minibatch B, the anchor set $\mathcal{Q}$ consists of four different subsets sharing the same shape $\mathrm{N_q} \times \mathrm{D}$: vessel pixel set $\mathcal{Q}_{VP}$, background pixel set $\mathcal{Q}_{BP}$, vessel region set $\mathcal{Q}_{VR}$, and background region set $\mathcal{Q}_{BR}$, where $\mathrm{N_q}$ is the length of each subset. Before sampling anchors, a projection head F_{proj} is first used to get the pixel embeddings of I: $X = F_{proj}(F_{cnn}(I)) \in \mathbb{R}^{\mathrm{D}\times\mathrm{H}\times\mathrm{W}}$, where D is the dimension of embeddings.

Then, pixels from Ω_P and points from Ω_R are sampled simultaneously, whose embeddings are used to fill these sets. Specifically, for an input image I, $\mathrm{N_{qi}}/2$ easy and $\mathrm{N_{qi}}/2$ hard pixels are pulled for $\mathcal{Q}_{VP}$, where misclassified pixel h_i with $\hat{Y}_{h_i}^v \leqslant \hat{Y}_{h_i}^b$ is treated as a hard one and pixel e_i with $\hat{Y}_{e_i}^v > \hat{Y}_{e_i}^b$ is an easy one. The same process is used for $\mathcal{Q}_{BP}$:

$$\begin{aligned} \mathcal{Q}_{VP} &= \mathcal{Q}_{VP} \cup \left\{q_{h_i}, q_{e_i} \in X | i = 1,2,..., \mathrm{N_{qi}}/2; h_i, e_i \in \Omega_P^{\mathrm{v}}\right\} \\ \mathcal{Q}_{BP} &= \mathcal{Q}_{BP} \cup \left\{q_{h_i}, q_{e_i} \in X | i = 1,2,..., \mathrm{N_{qi}}/2; h_i, e_i \in \Omega_P^{\mathrm{b}}\right\} \end{aligned} \tag{8}$$

where h means "hard" and e means "easy". Meanwhile, the region anchors of $\mathrm{N_{qi}}$ randomly sampled annotated vessel points and background points are calculated to fill $\mathcal{Q}_{VR}$ and $\mathcal{Q}_{BR}$, respectively:

$$\begin{aligned} \mathcal{Q}_{VR} &= \mathcal{Q}_{VR} \cup \{q_i | i = 1, 2, ..., \mathrm{N_{qi}}\} \\ \mathcal{Q}_{BR} &= \mathcal{Q}_{BR} \cup \{q_i | i = 1, 2, ..., \mathrm{N_{qi}}\} \end{aligned} \tag{9}$$

To emphasize wrong predictions in $\hat{Y}$, the region anchor q_i of a labeled point is obtained by weighted average of all pixel embeddings in this point:

$$q_i = \frac{1}{\left|\mathcal{R}_i^k\right|}\sum_{j \in \mathcal{R}_i^k}(1 - \hat{Y}_j^k)X_j \in \mathbb{R}^{\mathrm{D}} \tag{10}$$

where j is the pixel in this point and k is the ground truth class of pixel j. Above process is repeated for each image in B, and $\mathrm{N_q}$ equals $\mathrm{N_{qi}} \times \mathrm{B}$.

Moreover, with point annotations, vessel boundaries do not receive effective supervision. Therefore, ω_i is used to give more attention on these regions in $\mathcal{L}_{cl}$, as shown in Fig. 2(B). Formally, ω_i is defined as:

$$\omega_i = 1 - d_i \cdot \mathbb{1}(q_i \in \mathcal{Q}_{VP}) \tag{11}$$

where $\mathbb{1}$ denotes if q_i is from $\mathcal{Q}_{VP}$, and d_i is the distance of q_i from the center of the labeled point to which q_i belongs, normalized by the radius of this point.

2) Memory Bank Update: Memory bank can provide sufficient pos/neg samples which are important for contrastive learning [33], and formally, a vessel queue $\mathcal{K}_v \in \mathbb{R}^{\mathrm{N_k}\times\mathrm{D}}$ and a background queue $\mathcal{K}_{bg} \in \mathbb{R}^{\mathrm{N_k}\times\mathrm{D}}$ are used in PVN to build the memory bank $\mathcal{K}$, where $\mathrm{N_k}$ is the length of each queue.

Here, for I, a momentum encoder which shares the same structure as $\mathcal{M}_{seg}$ is used to get the consistent embeddings $X_{me} \in \mathbb{R}^{\mathrm{D}\times\mathrm{H}\times\mathrm{W}}$ and its predicted probability map is $\hat{Y}_{me} \in [0, 1]^{\mathrm{K}\times\mathrm{H}\times\mathrm{W}}$. Then, embeddings of $\mathrm{N_{ki}}$ randomly sampled annotated vessel pixels from Ω_P are added to $\mathcal{K}_v$, and the same number of labeled background pixel embeddings are added to $\mathcal{K}_{bg}$. For representing regions, $\mathrm{N_{ki}}$ labeled vessel points are

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selected from Ω_R and the averaged embeddings of pixels in each point are calculated and added to $\mathcal{K}_v$, and the similar process is used for $\mathcal{K}_{bg}$.

To get more representative features for vessels and background and to avoid misleading the training, all selected pixels and pixels used in generating region samples are filtered by a mask matrix to remove unconfident position:

$$M_j^k = \begin{cases} 1, & \widehat{Y}_{me,j}^k > 0.5 \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where j means each of the selected pixels. Meanwhile, the oldest $2\mathrm{N_{ki}}$ vessel samples and $2\mathrm{N_{ki}}$ background samples are dequeued.

D. Loss Function

With above designs, $\mathcal{M}_{seg}$ can benefit from vessel prior knowledge and generates improved representations. In all, given an input image I, the overall loss function is:

$$\mathcal{L} = \mathcal{L}_{pce} + \lambda_{area}\mathcal{L}_{area} + \lambda_{cl}\mathcal{L}_{cl} \tag{13}$$

where λ_{area} and λ_{cl} are used to balance different objectives.

IV. Experiments

A. Dataset

The LSCI fundus images dataset (PKU-LSCI) and 548 nm fundus images dataset (PKU-548) used in this study were collected by our custom-built multimodal fundus imaging system [3]. This system is equipped with a laser source with a wavelength of 850 nm for LSCI and the fundus images were taken with 548 nm illumination light. In all, PKU-LSCI and PKU-548 contains 165 and 340 images, respectively. Both datasets were randomly split on patient level for training (70%), validation (10%) and testing (20%). For PKU-LSCI and PKU-548, by using Pair software (Pair, Shenzhen, China), full masks were annotated for all images, and point annotations were randomly sampled from these masks for the training and validation set. The annotation time for Fig. 1(b) and (e) was estimated by manual annotation. All these manual annotations were qualified by ophthalmology experts. This study was conducted in accordance with the Declaration of Helsinki and was approved by the Ethics Committee of Peking University Health Science Center under PUIRB-YS2023166. Consent was given for publication by the participants.

Further evaluation was performed on three public datasets, including CHASE_DB1 [39], DRIVE [21] and HRF [40], where each dataset contains 28, 40, and 45 fundus images with the resolution of 999×960, 565×584, and 3504×2336 pixels, respectively. For these datasets, point annotations were randomly sampled from their full masks.

B. Implementation Details

UNet [36] was used as $\mathcal{M}_{seg}$ and $\mathcal{M}_{aux}$, and PAMs were generated from the features of the last two decoder layers in UNet. The $\mathcal{M}_{aux}$ was warmed for 20 epochs to get the initial estimation of vessel area and the momentum m for area update was set to 0.999. α^{upper} and α^{lower} are set to 1.1 and 0.9, respectively [14].

As for contrastive learning, the features before the segmentation head of $\mathcal{M}_{seg}$ were used as the input to the projection head F_{proj} which generated L2-normalized embeddings after two convolution layers. The dimension of all embeddings was 256. The length of each anchor set, $\mathrm{N_q}$, was 50. To update the memory bank, for each class in each image, $\mathrm{N_{ki}}$ was set to 20 and the length of memory banks, $\mathrm{N_k}$, was set to the product of the number of images in the training set and $2\mathrm{N_{ki}}$. Finally, for computing the infoNCE loss, 128 positive and 1024 negative samples were selected from the memory bank for each anchor, and τ was set to 0.07 [15].

All deep learning models were implemented in PyTorch, using GeForce RTX 4090 GPUs. PVN was trained using the Adam optimizer, of which the weight decay was set to 2×10^{-5}. The learning rate was set to 0.001 at first and gradually reduced. In the loss function, λ_{cl} was dynamically adjusted to ensure that the ratio between $\mathcal{L}_{pce}$ and $\mathcal{L}_{cl}$ is 20 and λ_{area} was set to 0.001. The best hyperparameters were selected based on the loss on the validation data. The minibatch size for training was set to 6 and the size of input images was $1\times512\times512$, and the loss of each minibatch is the average loss of all images in it. The maximum number of epochs was set to 600.

C. Evaluation

For comparisons and analysis of PVN, we adopted quantitative metrics including Dice coefficient (Dice), Jaccard similarity coefficient (JS), Cohen's kappa coefficient (K) and Accuracy (Acc). All of these metrics are presented in percentages, with higher values indicating better performance. All experiments were repeated across: (1) different network initialization and minibatch sampling, (2) different splits of training/validation/testing sets, and (3) different samplings of point annotations, where each type of experiments was repeated five times with different random seeds. We reported the mean and standard deviation (SD) of the results. For statistical significance, student's t-test was utilized, with * and ** representing $\mathrm{p} < 0.05$ and $\mathrm{p} < 0.01$, respectively.

D. Results on PKU-LSCI dataset

To demonstrate the effectiveness of PVN, comparisons were first conducted on PKU-LSCI dataset. UNet trained by (1) with $a_k = 1$ was used as our baseline, denoted as $\mathrm{UNet_{PCE}}$. Meanwhile, a weighted UNet, denoted as $\mathrm{UNet_{WPCE}}$, with $a_k = 1 - |\Omega_P^k|/|\Omega_P|$ was also compared. Different types of methods were used for comparison, including unsupervised methods used in previous works [17], [18], methods for vessel segmentation [4], [37], and methods for point annotations [14], [22], [24], [28], [38]. Specifically, [17] used the masks from DRIVE dataset to generate synthetic LSCI images to build the "fully labeled" dataset for later training (denoted as $\mathrm{CycleGAN_{S\text{-}DRIVE}}$ where "S" stands for "Synthetic"), and we further trained it with our larger dataset PKU-548 (denoted as $\mathrm{CycleGAN_{S\text{-}PKU}}$). Additionally, the LSCI images-to-masks generator, from a CycleGAN trained with LSCI images and full labels of other datasets, was used as the segmentor [18],

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TABLE I
COMPARISON RESULTS ON PKU-LSCI, PKU-548, CHASE_DB1, DRIVE, AND HRF.

Methods	Sup.	PKU-LSCI				PKU-548				CHASE_DB1				DRIVE				HRF			
		Dice↑	JS↑	K↑	Acc↑	Dice↑	JS↑	K↑	Acc↑	Dice↑	JS↑	K↑	Acc↑	Dice↑	JS↑	K↑	Acc↑	Dice↑	JS↑	K↑	Acc↑
CycleGAN$_{\text{DRIVE}}$ [18]	$\mathcal{U}$	50.92 ±0.78	34.33 ±0.73	46.72 ±0.76	91.87 ±0.35	54.48 ±0.27	37.61 ±0.26	50.21 ±0.32	92.01 ±0.08	24.58 ±0.96	14.04 ±0.62	18.28 ±0.20	86.68 ±3.74	- -	- -	- -	- -	32.83 ±1.18	19.70 ±0.85	25.76 ±0.91	85.07 ±4.90
CycleGAN$_{\text{PKU}}$ [18]	$\mathcal{U}$	58.57 ±0.79	41.58 ±0.79	54.59 ±0.92	92.24 ±0.30	- -	- -	- -	- -	33.41 ±0.92	20.08 ±0.67	28.20 ±1.95	88.24 ±5.80	26.18 ±0.68	15.09 ±0.45	17.62 ±0.21	83.63 ±2.56	38.84 ±0.13	24.14 ±0.09	32.18 ±0.60	86.91 ±1.88
CycleGAN$_{\text{S-DRIVE}}$ [17]	$\mathcal{U}$	65.02 ±0.09	47.73 ±0.43	61.86 ±0.41	95.05 ±0.05	63.46 ±0.16	46.78 ±0.25	59.71 ±0.40	92.80 ±0.25	21.15 ±0.02	11.29 ±0.62	14.47 ±1.15	88.85 ±0.51	- -	- -	- -	- -	49.59 ±0.08	33.08 ±0.10	44.26 ±0.03	89.71 ±0.13
CycleGAN$_{\text{S-PKU}}$ [17]	$\mathcal{U}$	69.32 ±0.12	53.55 ±0.81	66.83 ±0.68	94.95 ±0.11	- -	- -	- -	- -	54.60 ±0.07	37.60 ±0.06	50.75 ±0.08	92.33 ±0.24	30.20 ±0.35	17.85 ±0.22	21.67 ±1.10	82.94 ±2.55	58.15 ±0.52	41.12 ±0.52	53.48 ±0.46	91.04 ±0.10
UNet$_{\text{Full}}$ [36]	$\mathcal{F}$	82.22 ±0.28	69.85 ±0.38	80.81 ±0.32	97.40 ±0.09	78.77 ±0.64	65.23 ±0.83	76.58 ±0.71	96.00 ±0.14	80.39 ±0.74	67.27 ±1.01	79.06 ±0.80	97.49 ±0.12	81.43 ±0.29	68.71 ±0.40	79.66 ±0.33	96.76 ±0.08	79.68 ±0.75	66.42 ±1.02	78.02 ±0.78	96.91 ±0.07
DSCNet$_{\text{Full}}$ [4]	$\mathcal{F}$	81.80 ±0.39	69.25 ±0.55	80.31 ±0.40	97.23 ±0.05	78.27 ±0.60	64.54 ±0.77	75.99 ±0.66	95.86 ±0.14	79.09 ±1.31	65.52 ±1.67	77.61 ±1.41	97.20 ±0.21	81.05 ±0.41	68.17 ±0.56	79.23 ±0.45	96.67 ±0.08	79.07 ±0.74	65.60 ±1.01	77.34 ±0.77	96.79 ±0.08
Rolling-Unet$_{\text{Full}}$ [37]	$\mathcal{F}$	82.34 ±0.38	70.02 ±0.53	80.93 ±0.42	97.39 ±0.11	78.69 ±0.65	65.13 ±0.85	76.49 ±0.74	95.99 ±0.18	80.54 ±0.98	67.48 ±1.32	79.20 ±1.05	97.50 ±0.15	81.54 ±0.28	68.87 ±0.39	79.77 ±0.32	96.76 ±0.07	79.70 ±0.74	66.45 ±1.01	78.03 ±0.75	96.91 ±0.05
UNet$_{\text{PCE}}$ [36]	$\mathcal{P}$	70.28 ±1.23	54.24 ±1.46	67.42 ±1.42	94.46 ±0.42	70.92 ±0.78	55.10 ±0.94	67.34 ±0.88	93.36 ±0.23	68.97 ±0.49	52.76 ±0.57	66.48 ±0.58	95.16 ±0.28	74.24 ±1.01	59.08 ±1.27	71.40 ±1.21	94.74 ±0.43	70.74 ±0.57	54.96 ±0.69	67.79 ±0.65	94.44 ±0.25
UNet$_{\text{WPCE}}$ [36]	$\mathcal{P}$	68.41 ±0.71	52.04 ±0.81	65.30 ±0.81	93.91 ±0.26	68.79 ±0.57	52.59 ±0.65	64.77 ±0.66	92.45 ±0.25	65.26 ±1.26	48.55 ±1.37	62.32 ±1.43	94.12 ±0.47	72.12 ±0.53	56.46 ±0.66	68.99 ±0.75	94.15 ±0.57	69.07 ±1.09	53.01 ±1.26	65.87 ±1.35	93.89 ±0.62
DSCNet$_{\text{PCE}}$ [4]	$\mathcal{P}$	65.94 ±1.65	49.29 ±1.83	62.50 ±1.88	93.10 ±0.58	69.34 ±0.15	53.28 ±0.17	65.53 ±0.17	92.90 ±0.11	66.13 ±0.82	49.46 ±0.91	63.30 ±0.84	94.38 ±0.13	71.23 ±0.36	55.37 ±0.43	67.90 ±0.41	93.77 ±0.14	70.78 ±0.78	55.05 ±0.94	67.83 ±0.89	94.44 ±0.24
Rolling-Unet$_{\text{PCE}}$ [37]	$\mathcal{P}$	71.81 ±1.54	56.08 ±1.87	69.22 ±1.74	95.09 ±0.44	71.81 ±0.87	56.17 ±1.04	68.47 ±1.03	93.87 ±0.38	70.24 ±0.94	54.24 ±1.10	67.90 ±1.05	95.48 ±0.26	75.13 ±0.70	60.20 ±0.89	72.43 ±0.80	95.05 ±0.20	72.05 ±0.61	56.58 ±0.75	69.28 ±0.70	94.81 ±0.21
WESUP [22]	$\mathcal{P}$	56.99 ±0.60	39.91 ±0.59	52.32 ±0.69	90.07 ±0.25	48.51 ±0.46	32.14 ±0.42	40.42 ±0.60	82.22 ±0.43	47.98 ±0.58	31.62 ±0.50	42.66 ±0.58	87.59 ±0.28	51.17 ±1.51	34.46 ±1.35	44.45 ±1.57	85.93 ±0.38	44.61 ±0.84	28.77 ±0.69	37.87 ±0.47	85.56 ±1.90
SizeLoss [14]	$\mathcal{P}$	69.10 ±2.22	52.87 ±2.59	66.06 ±2.52	94.03 ±0.67	70.51 ±0.78	54.60 ±0.93	66.89 ±0.87	93.31 ±0.21	69.21 ±0.68	53.04 ±0.79	66.76 ±0.77	95.24 ±0.21	73.73 ±0.55	58.43 ±0.68	70.80 ±0.63	94.57 ±0.20	71.99 ±0.59	56.46 ±0.72	69.27 ±0.69	94.93 ±0.27
DenseCRF [28]	$\mathcal{P}$	71.52 ±1.86	55.75 ±2.27	68.80 ±2.12	94.75 ±0.56	71.59 ±0.34	55.90 ±0.40	68.14 ±0.37	93.66 ±0.14	70.77 ±0.58	54.83 ±0.69	68.45 ±0.64	95.50 ±0.15	75.46 ±0.59	60.62 ±0.76	72.81 ±0.73	95.14 ±0.28	73.09 ±0.53	57.84 ±0.65	70.46 ±0.60	95.10 ±0.11
AGMM [38]	$\mathcal{P}$	69.56 ±2.44	53.43 ±2.88	66.62 ±2.77	94.27 ±0.73	71.23 ±0.75	55.47 ±0.89	67.69 ±0.84	93.45 ±0.23	70.67 ±0.22	54.68 ±0.26	68.33 ±0.23	95.45 ±0.08	75.32 ±0.63	60.45 ±0.80	72.57 ±0.74	94.92 ±0.24	72.63 ±0.81	57.28 ±1.01	69.95 ±0.94	94.99 ±0.28
TEL [24]	$\mathcal{P}$	72.45 ±1.74	56.87 ±2.12	69.87 ±2.00	95.05 ±0.59	72.01 ±0.48	56.45 ±0.55	68.74 ±0.51	94.02 ±0.09	72.73 ±0.40	57.18 ±0.49	70.66 ±0.47	96.05 ±0.20	76.89 ±0.23	62.48 ±0.30	74.50 ±0.29	95.63 ±0.13	72.82 ±0.82	57.49 ±1.00	70.16 ±0.95	95.04 ±0.27
PVN(ours)**	$\mathcal{P}$	**80.87 ±0.38**	**67.90 ±0.53**	**79.31 ±0.42**	**97.11 ±0.08**	**75.65 ±0.32**	**60.99 ±0.41**	**73.08 ±0.33**	**95.33 ±0.09**	**74.71 ±0.57**	**59.69 ±0.74**	**73.03 ±0.58**	**96.84 ±0.12**	**77.59 ±0.39**	**63.41 ±0.52**	**75.48 ±0.41**	**96.13 ±0.07**	**74.28 ±0.22**	**59.29 ±0.29**	**72.16 ±0.23**	**96.07 ±0.04**

$\mathcal{U}$, $\mathcal{F}$, and $\mathcal{P}$ represent unpaired masks, fully supervised learning, and point-supervised learning, respectively.
Scores in **bold** style indicate the best, excluding the fully-supervised scores.
**: PVN has statistical significance over all other methods across all metrics, with $\mathrm{P} < 0.01$.

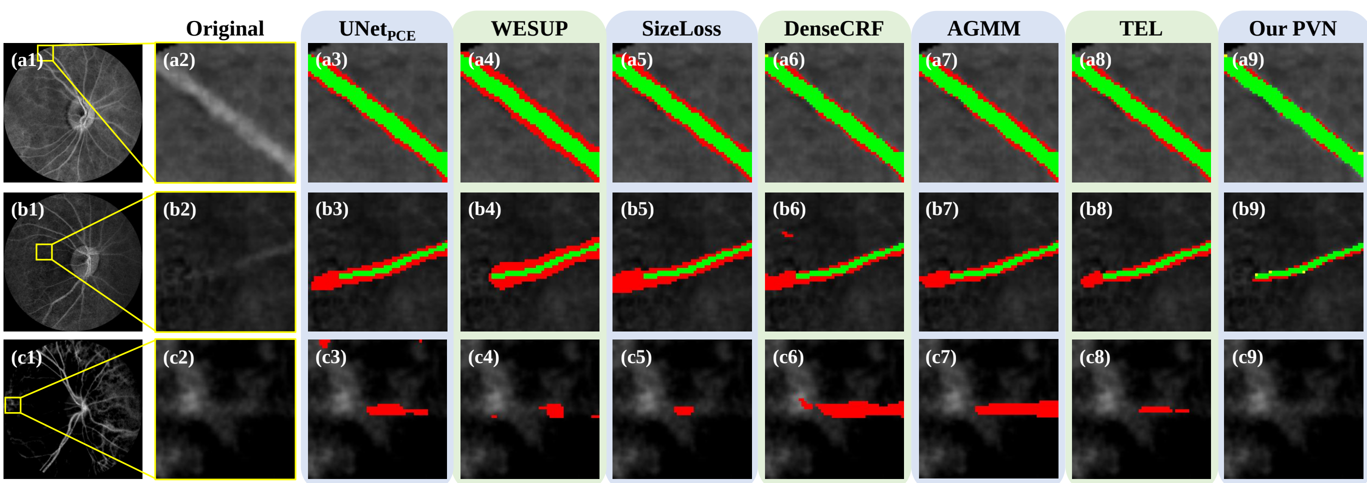

Fig. 4. Visual comparisons of different methods on PKU-LSCI Dataset: (a1)-(c1) Full-scale LSCI images from which the cropped images (a2)-(c2) are extracted (indicated by the yellow rectangles). Green: true positives, Red: false positives, Yellow: false negatives.

where masks from DRIVE dataset and our PKU-548 dataset were used, denoted as CycleGAN$_{\text{DRIVE}}$ and CycleGAN$_{\text{PKU}}$, respectively. As for SizeLoss, under weak supervision, it learns from both point labels and image tags bounds [14]. Fully-supervised networks (denoted with subscript "Full") were also trained with cross-entropy loss as the upper bound.

Table. I shows the quantitative results on PKU-LSCI dataset. Overall, our PVN outperformed other methods on all metrics,

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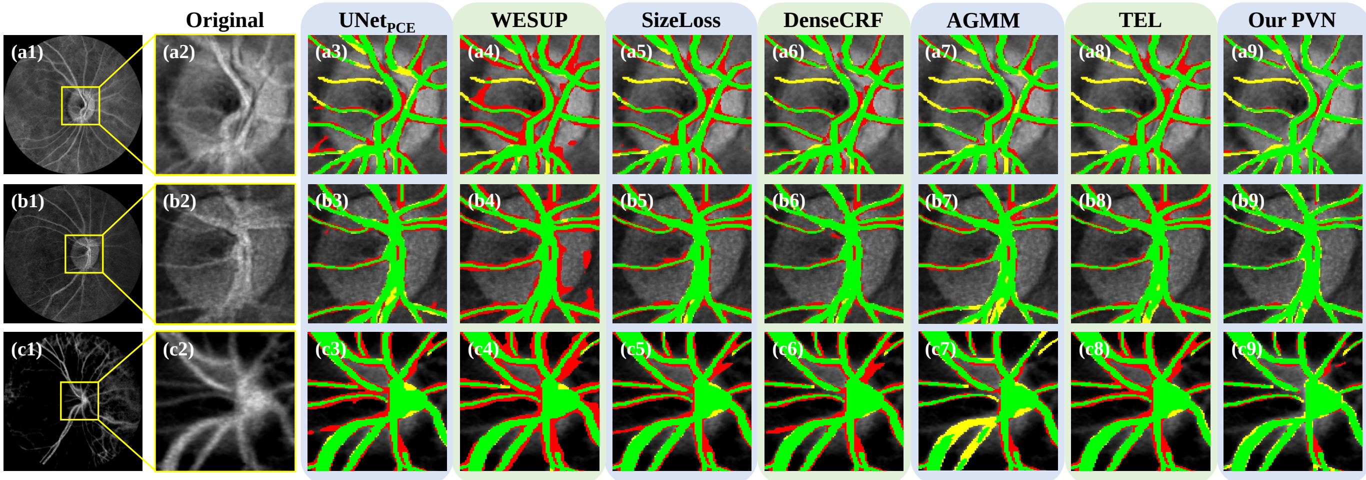

Fig. 5. Visual comparisons at the edges of the optic disc on PKU-LSCI Dataset: (a1)-(c1) Full-scale LSCI images from which the cropped images (a2)-(c2) are extracted (indicated by the yellow rectangles). Green: true positives, Red: false positives, Yellow: false negatives.

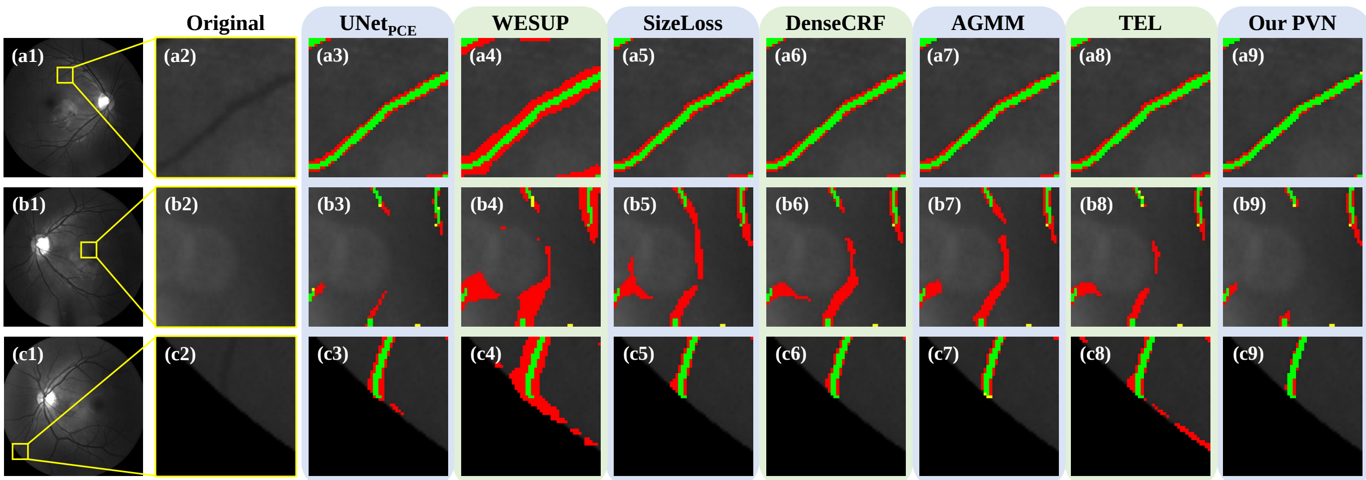

Fig. 6. Visual comparisons of different methods on PKU-548 Dataset: (a1)-(c1) Full-scale 548 nm fundus images from which the cropped images (a2)-(c2) are extracted (indicated by the yellow rectangles). Green: true positives, Red: false positives, Yellow: false negatives.

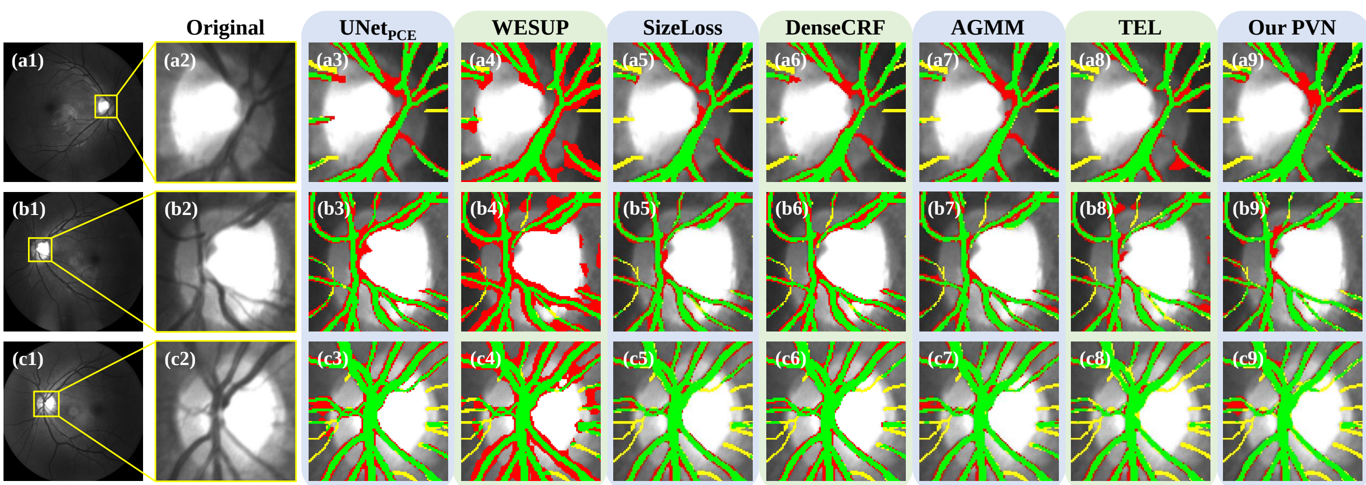

Fig. 7. Visual comparisons at the edges of the optic disc on PKU-548 Dataset: (a1)-(c1) Full-scale LSCI images from which the cropped images (a2)-(c2) are extracted (indicated by the yellow rectangles). Green: true positives, Red: false positives, Yellow: false negatives.

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with 80.87% Dice, 67.90% JS, 79.31% K and 97.11% Acc. Notably, after being incorporated into PVN, the baseline UNet_{PCE} obtained impressive improvements of 10.59% Dice, 13.66% JS, 11.89% K and 2.65% Acc. Different from fully-supervised tasks, the weighted partial cross-entropy loss did not take effect, which may come from that the difference in the number of annotated pixels of different classes is not significant. Although a larger dataset can improve performance, the CycleGAN-based methods showed relatively poor results, primarily due to the difficulty of generating realistic LSCI images and the lack of supervision in training the generator for the style transfer from LSCI images to binary masks. For the stability, as demonstrated by the SD values in Table. I, PVN is stable across various randomness including different network initialization and minibatch sampling, different dataset splits, and different samplings of point labels.

In addition to the quantitative results, Fig. 4 shows typical segmentation examples of our PVN compared with other methods, where PVN made fewer false positive predictions of both vessels at the image edges (Fig. 4(a1)-(a9)) and tiny vessels in low-contrast regions (Fig. 4(b1)-(b9)). At the same time, as shown in Fig. 4(c1)-(c2), the segmentation of vessels in LSCI images is challenging due to strong noise from the deep retina. All comparison methods classified the noisy region as vessels to some extent. In comparison, PVN showed strong robustness against the noise, as illustrated in Fig. 4(c3)-(c9). As for the vessels around the optic disc in Fig. 5, since most of the imaging light experiences total internal reflection through the axonal fibers and reflects off the capillaries on the disc surface, the optic disc is brighter, which makes the contrast between the vessels and the background different from other regions of the image. Thus, the segmentation performance degraded around the edges of the optic disc, which may be further exacerbated by less supervision around this region under point-supervised learning.

Overall, above results validate the feasibility of using point annotations for vessel segmentation and show the effectiveness of PVN.

E. Results on PKU-548 dataset

Experiments were also conducted on PKU-548 dataset, where the same comparison settings were used as in the previous section.

Table. I reports the results on PKU-548 dataset. Our PVN also achieved the best performance among other methods, with a Dice, JS, K and Acc of 75.65%, 60.99%, 73.08%, 95.33%, respectively. An improvement of 4.73% in Dice for the baseline UNet_{PCE} was achieved. This enhancement may not seem so remarkable compared with the results on PKU-LSCI dataset, since images in PKU-548 dataset contain many dense and small vessels, which increases the segmentation difficulty. Compared with the upper bound, although PVN was 3.12% behind in Dice, the point annotations save nearly 93% of the labeling time, which significantly reduces the labeling cost while obtaining results comparable to the fully-supervised method.

Some visual examples of PKU-548 dataset are illustrated in Fig. 6. It can be seen that PVN showed better performance for blurred vessels on low-contrast regions (Fig. 6(a1)-(a9)). PVN was also more robust to variations of image intensity and can obtain good results in the segmentation of small vessels near the macula (Fig. 6(b1)-(b9)). Moreover, as shown in Fig. 6(c1)-(c9), PVN was less affected by the image boundaries. However, affected by the sharp edges of the optic disc, poorer performance of all methods is also observed in PKU-548, as shown in Fig. 7.

F. Results on CHASE_DB1, DRIVE, and HRF

To further compare and demonstrate the performance of PVN, comparisons were conducted on public fundus datasets including CHASE_DB1, DRIVE, and HRF, and the same comparison settings were used, where methods denoted with subscript “Full” were trained with full annotated vessel labels. Results are reported in Table. I. Again, our PVN achieved the best results on all datasets with a significant improvement compared to the baseline method.

V. Discussion

1) Ablation studies: To prove the design behind PVN, ablation studies were conducted on the components of soft supervision and contrastive learning. Results are shown in Table. II as Mean_{SD}, where the performance on PKU-LSCI dataset and PKU-548 dataset are reported.

By removing $\mathcal{M}_{aux}$, we used PAMs generated from the last two layers in $\mathcal{M}_{seg}$ for area estimation, which showed significant drops of 19.07% and 2.9% in Dice compared to the full PVN with $\mathcal{M}_{aux}$. This is because $\mathcal{M}_{seg}$ converges slowly with both $\mathcal{L}_{area}$ and $\mathcal{L}_{cl}$ and thus generates area deviating from the real values, which would mislead the training. Moreover, since the 850 nm wavelength used in LSCI allows for deeper tissue penetration, it also captures deeper vessels, whose signals are usually weak and are considered as noise. As a result, PKU-LSCI images rely more heavily on accurate area estimation, leading to a more significant performance drop. This result confirms the effectiveness of using $\mathcal{M}_{aux}$. Furthermore, compared with the baseline, when only equipped with the proposed soft supervision approach, improvements of 8.74% and 2.31% in Dice can be achieved on both datasets, respectively. Meanwhile, on top of contrastive learning, the area loss can achieve a considerable improvement. It stems from that using learned shape priors as soft supervision on the predicted vessel shape encourages PVN to develop an understanding of vessel morphology, which is beneficial to the segmentation. As a more stable strategy, the momentum area update can also help achieve better performance.

Regarding the contrastive learning, by gradually adding each component, continuous performance increases can be observed in Table. II. Compared with the vanilla version which only used pixels, incorporating region features and position-based weights gave additional 0.89% and 1.62% improvements in Dice. It indicates that our pixels-and-regions-mixed contrastive learning with the weighting methods specifically designed for point labels can help to fully leverage the annotation information and learn better representations, which in turn improves the results.

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TABLE II
ABLATION STUDIES ON PKU-LSCI DATASET AND PKU-548 DATASET.

Settings	$\mathcal{L}_{area}$			$\mathcal{L}_{cl}$			PKU-LSCI				PKU-548			
	w/o $\mathcal{M}_{aux}$	w/ $\mathcal{M}_{aux}$	Momentum	Pixels	Regions	Position weights	Dice↑	JS↑	K↑	Acc↑	Dice↑	JS↑	K↑	Acc↑
1		✓	✓				$79.02_{0.50}$	$65.35_{0.68}$	$77.34_{0.56}$	$96.88_{0.11}$	$73.23_{0.36}$	$57.97_{0.42}$	$70.57_{0.34}$	$95.13_{0.04}$
2		✓	✓	✓			$79.98_{0.13}$	$66.67_{0.19}$	$78.42_{0.14}$	$97.09_{0.03}$	$74.03_{0.40}$	$58.96_{0.49}$	$71.37_{0.43}$	$95.14_{0.08}$
3		✓	✓	✓	✓		$80.07_{0.16}$	$66.79_{0.22}$	$78.51_{0.17}$	$97.11_{0.05}$	$74.40_{0.33}$	$59.41_{0.41}$	$71.78_{0.37}$	$95.21_{0.07}$
4				✓	✓	✓	$71.46_{0.36}$	$55.64_{0.44}$	$68.81_{0.40}$	$94.94_{0.10}$	$71.51_{0.30}$	$55.79_{0.35}$	$68.02_{0.37}$	$93.56_{0.17}$
5	✓		✓	✓	✓	✓	$61.80_{2.00}$	$45.04_{1.98}$	$59.22_{2.05}$	$94.97_{0.46}$	$72.75_{0.77}$	$57.43_{0.90}$	$70.00_{0.81}$	$94.97_{0.14}$
6		✓		✓	✓	✓	$79.51_{0.46}$	$66.03_{0.63}$	$77.91_{0.50}$	$97.03_{0.10}$	$71.57_{0.83}$	$56.04_{0.98}$	$68.90_{0.81}$	$95.04_{0.03}$
PVN		✓	✓	✓	✓	✓	$\mathbf{80.87_{0.38}}$	$\mathbf{67.90_{0.53}}$	$\mathbf{79.31_{0.42}}$	$\mathbf{97.11_{0.08}}$	$\mathbf{75.65_{0.32}}$	$\mathbf{60.99_{0.41}}$	$\mathbf{73.08_{0.33}}$	$\mathbf{95.33_{0.09}}$

TABLE III
STATISTICAL SIGNIFICANCE OF THE ABLATION STUDIES ON PKU-LSCI DATASET AND PKU-548 DATASET.

Setting pairs	PKU-LSCI				PKU-548			
	Dice	JS	K	Acc	Dice	JS	K	Acc
1, 2	$*$	$*$	$*$	$*$	$*$	$*$	$*$	
2, 3	$*$	$**$	$*$		$*$	$*$	$*$	
3, PVN	$*$	$*$	$*$		$**$	$**$	$**$	
4, PVN	$**$	$**$	$**$	$**$	$**$	$**$	$**$	$**$
5, PVN	$**$	$**$	$**$	$**$	$**$	$**$	$**$	$*$
6, PVN	$*$	$*$	$*$		$**$	$**$	$**$	$*$

$*$: P $<$ 0.05; $**$: P $<$ 0.01

As reported in Table. III, the statistical significance between different ablation settings was analyzed by student's t-test, where each of our designed modules achieves significant improvement.

2) Generalizations: To show the generalization of our method, we applied PVN to other segmentation backbones, including HRNet, DeepLabv3Plus, and Rolling-Unet. Specifically, for HRNet, the features from the highest resolution branch after stage4 and the features before the segmentation head were used in PAMs. As for DeepLabv3Plus, the features after the ASPP module and the features before the segmentation head were utilized. For Rolling-Unet, the features of the last two decoder layers were used. As results in Table. IV reflect, when incorporated into PVN, HRNet, DeepLabv3Plus, and Rolling-Unet obtain remarkable improvements on both datasets. These results demonstrate that PVN is effective and is easy to be applied to other networks.

Concerning the performance variation across different backbones, HRNet performs the parallel high-to-low resolution convolution streams along with cross-resolution feature fusion after a 4× downsampling of the inputs, which leads to the loss of image details. The decoder of DeepLabv3Plus lacks interaction with the high-resolution low-level features from the encoder, and it uses two 4× upsampling operations to recover the image resolution, where a more gradual process is preferred for fine-grained segmentation. In comparison, UNet has gradual downsamplings and upsamplings with the stride of two, and fully utilizes the low-level features through skip-connection for recovering more details, making it more suitable for segmenting objects with fine and tiny structures like vessels. As an improved version of UNet, Rolling-Unet utilizes MLPs to capture long-range dependency. When integrated into PVN, this more complex architecture may face increased training difficulty under multiple loss functions, which could account for the subtle performance differences compared with the UNet-based PVN.

3) Why not estimate vessel from the predicted probability?: For estimating area, intuitively, the probability map of vessels might be the first choice. However, as illustrated in Fig. 8(a4) and (b4), in the point-supervised scenario, the segmentation head tends to produce wider predictions with high confidence compared with the real vessel boundaries delineated in black, which would mislead the training. In comparison, PAM's activated regions in Fig. 8(a2)-(a3) and (b2)-(b3) align more closely with real vessels.

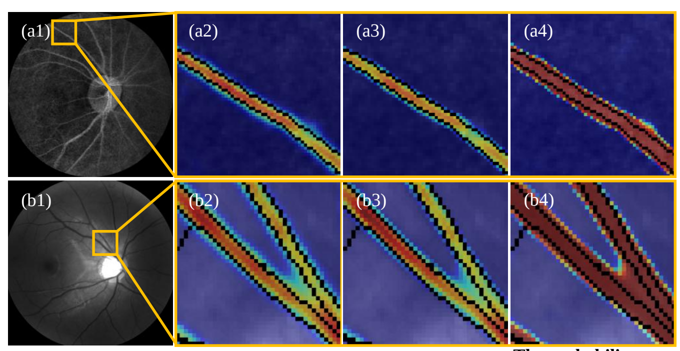

Fig. 8. Visualization of PAMs from different UNet decoder layers and the predicted probability map of vessels: (a1)-(b1) Full-scale LSCI and 548 nm fundus images from which the cropped images (a2)-(a4) and (b2)-(b4) are extracted (indicated by the yellow rectangles). (a2) and (b2) PAMs generated by the features of the second-to-last decoder layer in UNet, denoted as PAM_{up3}. (a3) and (b3) PAMs generated by the features of the last decoder layer in UNet, denoted as PAM_{up4}. (a4) and (b4) The predicted probability maps from UNet. Black boundaries: ground truth vessel boundaries.

To further demonstrate the effects, we estimated the vessel area from the probability map of vessels generated by $\mathcal{M}_{aux}$ in PVN for $\mathcal{L}_{area}$ and the results are shown in Table. V. Compared with the PAMs-based PVN, significant drops in performance can be observed, which quantitatively verify our choice of multilayer PAMs.

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TABLE IV

SEGMENTATION RESULTS BY ADDING PVN INTO OTHER SEGMENTATION BACKBONES.

Methods		Metrics on PKU-LSCI				Metrics on PKU-548			
		Dice↑	JS↑	K↑	Acc↑	Dice↑	JS↑	K↑	Acc↑
HRNet	PCE	$66.59_{2.09}$	$50.02_{2.36}$	$63.27_{2.39}$	$93.41_{0.72}$	$68.01_{0.49}$	$51.71_{0.51}$	$64.00_{0.52}$	$92.52_{0.18}$
	PVN	$\mathbf{78.61}^{\uparrow 12.01}_{0.67}$	$\mathbf{64.80}^{\uparrow 14.78}_{0.89}$	$\mathbf{76.87}^{\uparrow 13.60}_{0.75}$	$\mathbf{96.76}^{\uparrow 3.35}_{0.18}$	$\mathbf{69.13}^{\uparrow 1.12}_{0.27}$	$\mathbf{52.95}^{\uparrow 1.24}_{0.31}$	$\mathbf{65.85}^{\uparrow 1.85}_{0.27}$	$\mathbf{94.04}^{\uparrow 1.52}_{0.06}$
	Full	$80.99_{0.96}$	$68.11_{1.33}$	$79.47_{1.03}$	$97.18_{0.16}$	$71.76_{1.08}$	$56.11_{1.31}$	$68.95_{1.14}$	$94.83_{0.14}$
DeepLabv3Plus	PCE	$63.27_{2.05}$	$46.37_{2.21}$	$59.46_{2.37}$	$92.24_{0.80}$	$65.94_{0.45}$	$49.40_{0.48}$	$61.65_{0.43}$	$91.94_{0.34}$
	PVN	$\mathbf{70.27}^{\uparrow 7.00}_{1.56}$	$\mathbf{54.31}^{\uparrow 7.94}_{1.82}$	$\mathbf{67.96}^{\uparrow 8.49}_{1.68}$	$\mathbf{95.67}^{\uparrow 3.43}_{0.25}$	$\mathbf{68.66}^{\uparrow 2.71}_{0.20}$	$\mathbf{52.42}^{\uparrow 3.02}_{0.21}$	$\mathbf{65.38}^{\uparrow 3.73}_{0.16}$	$\mathbf{94.04}^{\uparrow 2.10}_{0.09}$
	Full	$79.49_{0.27}$	$66.01_{0.37}$	$77.83_{0.27}$	$96.91_{0.08}$	$71.94_{0.72}$	$56.41_{0.80}$	$68.96_{0.76}$	$94.57_{0.11}$
Rolling-Unet	PCE	$71.81_{1.54}$	$56.08_{1.87}$	$69.22_{1.74}$	$95.09_{0.44}$	$71.81_{0.87}$	$56.17_{1.04}$	$68.47_{1.03}$	$93.87_{0.38}$
	PVN	$\mathbf{79.88}^{\uparrow 8.08}_{0.67}$	$\mathbf{66.55}^{\uparrow 10.47}_{0.93}$	$\mathbf{78.31}^{\uparrow 9.08}_{0.72}$	$\mathbf{97.07}^{\uparrow 1.98}_{0.09}$	$\mathbf{74.65}^{\uparrow 2.84}_{0.18}$	$\mathbf{59.73}^{\uparrow 3.56}_{0.22}$	$\mathbf{72.04}^{\uparrow 3.57}_{0.17}$	$\mathbf{95.24}^{\uparrow 1.37}_{0.08}$
	Full	$82.34_{0.38}$	$70.02_{0.53}$	$80.93_{0.42}$	$97.39_{0.11}$	$78.69_{0.65}$	$65.13_{0.85}$	$76.49_{0.74}$	$95.99_{0.18}$

TABLE V

SEGMENTATION RESULTS OF PVN USING AREA ESTIMATED FROM THE PROBABILITY MAP OF VESSELS.

Dataset	Methods	Dice↑	JS↑	K↑	Acc↑
PKU-LSCI	The probability map	$67.70_{1.04}$	$51.26_{1.18}$	$64.59_{1.18}$	$93.94_{0.33}$
	PAMs	$\mathbf{80.87_{0.38}}$	$\mathbf{67.90_{0.53}}$	$\mathbf{79.31_{0.42}}$	$\mathbf{97.11_{0.08}}$
PKU-548	The probability map	$70.64_{0.50}$	$54.76_{0.61}$	$67.24_{0.65}$	$93.78_{0.37}$
	PAMs	$\mathbf{75.65_{0.32}}$	$\mathbf{60.99_{0.41}}$	$\mathbf{73.08_{0.33}}$	$\mathbf{95.33_{0.09}}$

4) Robustness to different percentages of labeled pixels: In this section, we analyze the robustness of PVN to different percentages of annotated vessel pixels. For this purpose, we sampled the point annotations in PKU-LSCI dataset and PKU-548 dataset to build datasets where the number of annotated vessel pixels in sparse points accounts for 1%, 3% and 5% of the total vessel pixels, respectively. For background, after sampling, the number of annotated background points is equal to the number of annotated vessel points. The UNet-based PVN and UNet_{PCE} were used in this analysis and the results are reported in Fig. 9 as Mean_{SD}.

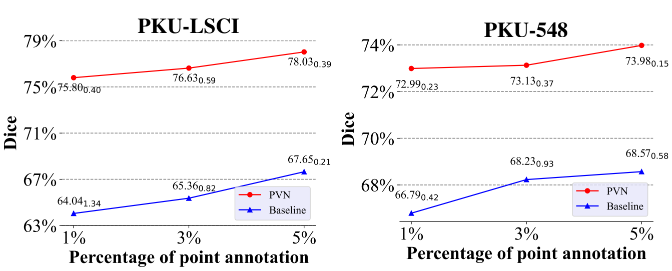

Fig. 9. Dice obtained by PVN and UNet_{PCE} when changing the percentages of the annotated pixels in PKU-LSCI dataset (left) and PKU-548 dataset (right). Results are presented as Mean_{SD}.

As shown in Fig. 9, remarkably, even with only 1% of vessel pixels annotated, PVN can also obtain impressive segmentation results, with the dice of 75.80% and 72.99% for PKU-LSCI and PKU-548, respectively. Furthermore, PVN shows strong robustness to varying percentages of labeled pixels from 5% to 1%, compared with the quick drops in performance of the baseline UNet_{PCE}.

5) Clinical advantages and applications: To further demonstrate the advantages of our method in real clinical scenarios, experiments on the evaluation of retinal oxygen kinetics were conducted, including arteriovenous oxygen saturation difference (SO_2^{va}), arteriovenous oxygen concentration difference (CO_2^{va}), arterial blood flow (RBF^a) and venous blood flow (RBF^v), all of which rely on accurate vessel segmentation. For this evaluation, five subjects were randomly selected from our dataset, and calculations were performed using the software of our custom-built multimodal fundus imaging system [3], with segmentation results of different models. Metrics derived from manual annotations serve as the ground truth with a SO_2^{va}, CO_2^{va}, RBF^a and RBF^v of 34.84±4.26 %, 6.71±1.12 mlO_2/dL, 12.74±0.36 $\mu L/min$, and 13.51±0.18 $\mu L/min$, respectively. Table. VI reports the errors between the results generated from different models and the ground truth, along with the standard deviation of these errors, where PVN achieves the smallest errors and significantly outperforms all other methods under the student's t-test. These results show that our method improves the evaluation precision under an efficient annotation strategy compared with other methods, which helps the applications in clinic.

TABLE VI

ERRORS OF DIFFERENT METHODS IN CLINICAL EVALUATIONS OF RETINAL OXYGEN KINETICS

Methods	Errors of Metrics (unit) ↓			
	SO_2^{va} (%)	CO_2^{va} (mlO_2/dL)	RBF^a ($\mu L/min$)	RBF^v ($\mu L/min$)
UNet_{PCE} [36]	8.82±6.71	2.45±1.31	11.20±3.22	10.05±0.73
WESUP [22]	12.93±11.57	9.74±2.46	43.59±10.22	37.55±3.12
SizeLoss [14]	5.03±4.18	1.44±0.98	10.47±2.13	8.13±0.62
DenseCRF [28]	4.69±2.17	1.87±0.37	10.32±2.04	8.90±0.09
AGMM [38]	7.53±4.29	1.85±0.96	5.42±1.46	7.12±0.48
TEL [24]	1.11±0.13	1.50±0.24	4.70±1.59	3.97±0.58
PVN(ours)*	**0.57±0.05**	**1.29±0.20**	**1.10±0.64**	**1.62±0.12**

*: PVN has statistical significance over all other methods across all metrics, with $P < 0.05$.

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For facilitating clinical applications, fundus vessel segmentation is a fundamental task which enables the quantification of vessel structure and promotes the diagnosis of ocular diseases such as diabetic retinopathy and glaucoma [41]. Additionally, since retinal vessels are closely related to cerebral vasculature and coronary arteries, vessel parameters can also improve the diagnosis of hypertension and stroke [42].

However, obtaining vessel segmentation is time-consuming, especially in certain clinical scenarios. Specifically, it is necessary to construct large datasets for new imaging modalities and multimodal diagnosis to obtain clinically applicable vessel segmentation models [43]–[45]. Besides, in large cohort studies on different diseases and novel therapies, annotation of new datasets is required to deal with new image appearance. Moreover, existing datasets also need to be continuously expanded to accommodate complex scenarios, such as images of rare diseases. For these situations, our study not only shows the effectiveness and efficiency of point supervision, but also proposes a more accurate segmentation approach, which easily offers vessel information and enhances clinical applications and patient care.

VI. Limitations

While PVN has excellent performance, certain limitations should also be considered. First, as shown in Fig. 10(a1)-(a2), abnormal strong reflections in certain regions of the retina cause high grayscale values in LSCI images and affect PVN's ability to identify vessels. Additionally, misalignment during imaging can lead to light leakage at the edges, which compromises PVN's performance (Fig. 10(b1)-(b2)). Moreover, PVN cannot effectively handle fundus images of unseen diseases, such as retinitis pigmentosa (Fig. 10(c1)-(c2)). Addressing these issues will be an important direction for our future improvements.

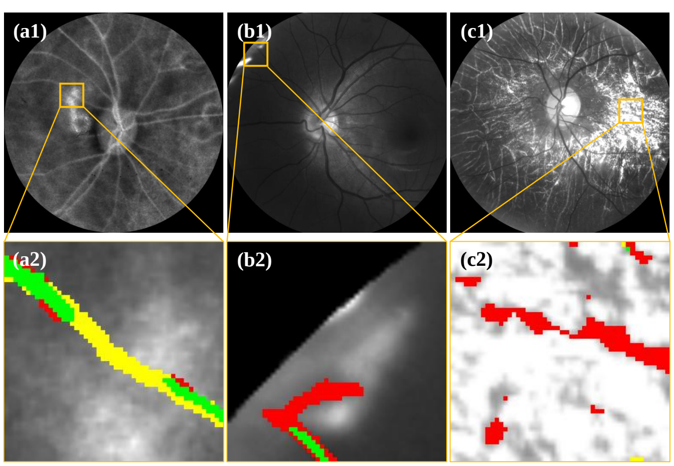

Fig. 10. Visualization of failure cases of PVN. (a1)-(c1)Full-scale images from which the cropped images (a2)-(c2) are extracted (indicated by the yellow rectangles).Green: true positives, Red: false positives, Yellow: false negatives.

To evaluate computational cost, GPU memory required for each method with a minibatch size of 6, as well as the FLOPs are reported in Table. VII, both under 512×512 input resolution. The evaluations were conducted using NVIDIA RTX 4090 GPUs and PyTorch 2.3.1 with single-precision and distributed data parallel. Since PVN contains several submodules, it has higher FLOPs and needs more GPU memory compared with some methods. However, after training, only $\mathcal{M}_{seg}$ is used for later test, and the computational cost at this stage is only that of the baseline. Designing more efficient method is an important aspect of our future work.

TABLE VII
Comparison Results of GPU Memory and FLOPs.

Methods	Training		Testing	
	GPU Mem (GiB†)	FLOPs (G‡)	GPU Mem (GiB†)	FLOPs (G‡)
CycleGAN [20]	9.57	762.57	0.97	47.66
UNet$_{\text{PCE}}$ [36]	12.11	320.72	1.19	319.80
DSCNet [4]	138.43	121.27	13.53	121.13
Rolling-Unet [37]	11.47	262.89	1.12	262.23
WESUP [22]	121.30	4935.29	11.28	1873.50
SizeLoss [14]	12.13	320.72	1.19	319.80
DenseCRF [28]	12.11	320.16	1.19	319.80
AGMM [38]	14.74	320.86	1.19	319.80
TEL [24]	27.32	330.11	1.19	319.80
PVN(ours)	20.14	1304.10	1.19	319.80

† *1 GiB=2^{30} bytes.*
‡ *G means 10^9.*

VII. Conclusion

In this paper, we introduce the use of point annotations into fundus vessel segmentation and design a new method for it. Our method generates soft supervision from multilayer-based PAMs and fully leverages the labeled points by using our pixels-and-regions-mixed contrastive learning with weighting methods specifically designed for point annotations. Experiments verified the performance of our method and indicated that our method can generalize to various segmentation models. Even with 1% of total vessel pixels annotated, our method can still achieve excellent performance.

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